

De-multiplexing of Address/Data Bus (Self-Learning)

Q. Why does the 8086 use a multiplexed Address/Data bus? Explain the process of de-multiplexing this bus using Address Latch Enable (ALE) and Latches.

✦ Definition to Remember

To save physical pins on the 40-pin 8086 IC, Intel used a **Multiplexed Bus** (AD0-AD15), where the same pins carry the Address during clock cycle T1, and Data during T2-T4. **De-multiplexing** is the hardware process of using external **Latches** and the **ALE** signal to extract and hold the address before the pins switch to data mode.

1. Need for Multiplexing and De-multiplexing

- **Multiplexing:** The 8086 needs 20 pins for the address and 16 pins for data. Having 36 separate pins on a 40-pin chip leaves no room for power, clock, and control signals. Hence, they are multiplexed.
- **De-multiplexing:** Memory chips require a stable, constant address for the entire read/write cycle. If the 8086 removes the address after T1 to send data, memory loses track of the location. The address must be externally locked in place (latched).

2. The Role of ALE (Address Latch Enable)

- The 8086 generates a positive pulse on the **ALE** pin strictly during the **T1 state** of every machine cycle.
- This pulse guarantees that the signals currently on the AD pins represent a valid memory address.

3. Using Latches (74LS373) for De-multiplexing

To de-multiplex, external **D-type transparent latches** (e.g., 74LS373) are used.

* Hardware Connection:

- * The AD bus (from CPU) connects to the Latch Inputs.
- * The ALE pin (from CPU) connects to the **Latch Enable (LE)** pin.

* Working Mechanism:

1. **During T1:** 8086 puts the Address on the bus and pulses ALE High.
2. Because ALE is High, the latches become "transparent," passing the address through to their outputs.
3. **End of T1:** ALE goes Low. The latches "lock" (latch) the address present at their inputs.
4. **During T2, T3, T4:** The 8086 removes the address and uses the AD lines for Data. However, the memory safely receives a constant address from the outputs of the latches.

★ **Must-Write Points (for 10 marks)**

1. **Multiplexing** saves pins (AD0-AD15 act as Address in T1, Data in T2-T4).
2. Memory requires a stable address for the entire cycle, creating the need for **De-multiplexing**.
3. De-multiplexing separates the Address and Data using external **Latches** (e.g., 74LS373).
4. The **ALE (Address Latch Enable)** signal is pulsed High by the CPU only during **T1**.
5. ALE connects to the Enable pin of the external latches.
6. When ALE is High (T1), the latches grab the address from the AD bus.
7. When ALE goes Low, the latches lock the address and hold it steady for the memory, while the CPU uses the AD bus for data in T2-T4.

⚡ **Quick Recall**

Multiplexing saves pins → Memory needs stable address → Use Latches (74LS373) + ALE signal → T1: ALE is High, Latches grab Address → T2-T4: ALE Low, Latches hold Address, Bus used for Data